

Foundations of Research

1. Introduction to Research

- 1.1 Meaning of Research
- The word research is derived from the French word recherche, meaning “to search again” or “to investigate carefully.”
- In an academic context, research is a systematic, objective, and scientific investigation undertaken to discover new knowledge, verify existing knowledge, solve problems, establish relationships, or develop theories and explanations.
- In simple terms: Research is a systematic process of asking questions, collecting relevant evidence, analyzing that evidence, and arriving at justified conclusions.

- Research is more than simply collecting information.
- It involves:

Problem → Question → Evidence → Analysis → Interpretation → Conclusion → Knowledge

- Example
 - Suppose a university wants to know: "Does the use of generative AI improve students' programming performance?"
 - Simply asking students whether they like ChatGPT is not sufficient research.

- A research study may involve:
 - Defining the research problem.
 - Formulating research questions.
 - Reviewing previous literature.
 - Developing hypotheses.
 - Selecting students as a sample.
 - Designing an experiment.
 - Measuring programming performance.
 - Collecting data.
 - Applying statistical tests.
 - Interpreting the findings.
 - Drawing conclusions.
 - Identifying limitations and future research.
- Thus, research is a **structured investigation**, rather than an informal search for information.

2. Definitions of Research

- General definition: Research is a systematic investigation designed to establish facts, develop principles, or generate new knowledge.
- Scientific perspective: Research can be understood as:
 - A systematic and controlled investigation of a problem through observation, measurement, experimentation, analysis, and interpretation.
- Important elements common to definitions:
 - Almost all definitions of research contain some combination of:
 - Systematic investigation
 - Problem identification
 - Collection of evidence
 - Analysis
 - Interpretation
 - Objectivity
 - Logical reasoning
 - Verification
 - Generation of knowledge
 - Problem solving

3. What Research Is and What Research Is Not

- **Research IS:**
 - Systematic
 - Planned
 - Logical
 - Evidence-based
 - Objective
 - Analytical
 - Reproducible where appropriate
 - Ethical
 - Critical
 - Purpose-oriented
- **Research IS NOT:**
 - Random information gathering
 - Mere reading of books
 - Copying information from the Internet
 - Personal opinion
 - Guesswork
 - Unverified assumptions
 - Data collection without a research question
 - Manipulation of evidence to support a desired conclusion

- Important distinction
 - Information $\neq$ Knowledge $\neq$ Research
 - For example: "Students who practice more programming tend to score higher."
 - This is information/observation.
 - Research investigates: "Is there a statistically significant relationship between programming practice time and programming examination performance?"

4. Nature of Research

- 4.1 Systematic:
 - Research follows an organized sequence of activities.
 - For example: Problem identification → Literature review → Research questions → Methodology → Data collection → Analysis → Conclusion
- 4.2 Logical:
 - Research should follow sound reasoning.
 - Conclusions must follow logically from evidence.
- 4.3 Empirical:
 - Where appropriate, research relies on observable and measurable evidence.
 - Examples:
 - Examination scores
 - Survey responses
 - Experimental observations
 - Sensor readings
 - Interview transcripts
 - Musical features
 - Biological measurements

- 4.4 Objective:
 - Researchers should minimize personal bias.
 - For example, a researcher should not reject data merely because it contradicts their hypothesis.
- 4.5 Controlled:
 - Where possible, researchers control relevant variables to determine cause-and-effect relationships.
 - This is particularly important in experimental research.
- 4.6 Replicable:
 - A research study should provide enough methodological information for other researchers to understand and, where feasible, reproduce the study.
- 4.7 Critical:
 - Researchers must critically evaluate:
 - Assumptions
 - Data
 - Methods
 - Previous research
 - Results
 - Conclusions
 - Limitations
- 4.8 Ethical:
 - Research must respect:
 - Participants
 - Privacy
 - Confidentiality
 - Intellectual property
 - Academic integrity
 - Informed consent
 - Research integrity

5. Objectives of Research

- The objectives of research depend on the nature of the study.
- The major objectives are:
 - Exploration
 - Description
 - Explanation
 - Prediction
 - Control
 - Problem solving
 - Theory development
 - Theory testing
 - Knowledge generation
 - Knowledge verification

- 5.1 Exploratory Objective

- Exploratory research is conducted when relatively little is known about a problem.
- PurposeTo explore:
 - What is happening?
 - Why might it be happening?
 - What variables may be relevant?
 - What questions should be investigated?
- Example
 - Exploring how university teachers perceive the use of generative AI in teaching.
 - The researcher may initially conduct interviews with teachers to identify important themes.

- 5.2 Descriptive Objective:
- Descriptive research aims to accurately describe a phenomenon, population, situation, or condition.
- It answers:
 - What?
 - Who?
 - Where?
 - When?
 - How much?
 - How often?
- Example
 - What percentage of postgraduate students use generative AI tools for academic work?
 - The objective is description rather than determining causation.

- 5.3 Explanatory Objective

- Explanatory research attempts to explain why or how something happens.
- It investigates relationships and possible causal mechanisms.
- Example
 - Why does the use of an adaptive learning system improve programming performance?

- 5.4 Predictive Objective

- Research can be used to predict future or unknown outcomes.
- Example
 - Using historical student data to predict:
 - Which students are at risk of failing a programming course?
 - Machine learning research frequently has a strong predictive objective.

- 5.5 Control Objective:

- Some research aims to determine how a phenomenon can be controlled or influenced.
- Example
 - Determining which teaching intervention most effectively improves student performance.

- 5.6 Problem-Solving Objective:

- Applied research often aims directly at solving practical problems.
- Example
 - Developing an AI-based system for identifying students requiring academic intervention.

- 5.7 Theory Development:

- Research may generate concepts, relationships, models, or theories.
- For example:
 - Observation: Certain musical patterns repeatedly occur in performances associated with a particular raga.
 - Research may lead to a theoretical model describing relationships between:
 - Raga → Musical structure → Perceived emotion

- 5.8 Theory Testing:

- Existing theories can be tested against empirical evidence.
- Example:
 - Testing whether a previously proposed learning theory applies to postgraduate programming education.

6. Major Purposes of Research

- Research can broadly serve four major purposes:

Purpose	Main Question
• Exploration	What is happening?
• Description	What does it look like?
• Explanation	Why/how does it happen?
• Prediction	What is likely to happen?

- A single research project can have multiple purposes.

7. Motivation for Research

- 7.1 Meaning of Research Motivation
 - Research motivation refers to the reasons that encourage a researcher to investigate a particular problem or question.
 - In other words:
 - Research motivation is the intellectual, practical, social, professional, or personal reason that drives a researcher to conduct a study.
 - A research problem should not be selected merely because it is interesting. There should be a meaningful justification for investigating it.

8. Sources of Research Motivation

- 8.1 Curiosity

- Researchers often begin with a question:
 - "Why does this happen?" or "Can this be done differently?"
- Curiosity is one of the fundamental drivers of scientific discovery.

- 8.2 Knowledge Gap

- A knowledge gap occurs when existing research does not adequately answer a particular question.
- Example
 - Existing research may study:
 - Western music and emotion.
 - But relatively little research may examine:
 - Computational modeling of emotional characteristics of Indian Classical Music.
- This gap can motivate research.

- 8.3 Practical Problems

- Real-world problems are major sources of research motivation.
- Examples:
 - Healthcare challenges
 - Educational difficulties
 - Cybersecurity threats
 - Environmental problems
 - Business inefficiencies
 - Data privacy issues

- 8.4 Contradictory Findings:

- Research findings sometimes disagree.
- For example:
 - Study A finds a positive relationship.
 - Study B finds no relationship.
 - Study C finds a negative relationship.
 - This contradiction creates an opportunity for further research.

- 8.5 Replication
 - A researcher may repeat an earlier study to determine whether its findings remain valid under:
 - Different conditions
 - Different populations
 - Different geographical regions
 - Different time periods
 - Different methodologies
- 8.6 Personal or Professional Interest
 - Researchers may investigate topics related to their academic specialization or professional experience.
 - However, personal interest should not compromise objectivity.
- 8.7 Social Need
 - Research may be motivated by the need to address social problems.
 - Examples:
 - Digital divide
 - Educational inequality
 - Poverty
 - Accessibility
 - Public health
 - Sustainable development

- 8.8 Technological Development

- New technologies frequently create new research questions.
- Examples:
 - Artificial Intelligence
 - Generative AI
 - Quantum computing
 - Internet of Things
 - Blockchain
 - Robotics

- 8.9 Policy and Decision-Making

- Governments and organizations need evidence to make decisions.
- Research can provide evidence for:
 - Policy formulation
 - Resource allocation
 - Educational planning
 - Healthcare planning
 - Economic decisions

9. Characteristics of Good Research Motivation

- A strong research motivation should be:
 - Relevant
 - Clearly stated
 - Scientifically meaningful
 - Practically significant where applicable
 - Supported by existing literature
 - Feasible
 - Ethical
 - Specific
- Weak motivation
 - "I selected this topic because it is interesting."
- Strong motivation
 - "Existing studies have examined AI-assisted programming education primarily among undergraduate students, while evidence regarding its effectiveness for postgraduate learners remains limited. This gap motivates an empirical investigation."

10. Theory

- 10.1 Meaning of Theory

- A theory is a systematic and logically organized set of concepts, propositions, assumptions, and relationships that explains or predicts phenomena.
- In simple terms:
 - A theory is an organized explanation of how and why certain phenomena are related.
- A theory is not simply a guess.

11. Theory vs Common-Sense Explanation

Common Sense

Informal explanation

May rely on intuition

Often vague

Difficult to test

May be inconsistent

Usually lacks formal concepts

Theory

Systematic explanation

Based on evidence and reasoning

Clearly defined

Testable where empirical

Logically organized

Uses concepts and relationships

- **Example**

- Common statement:
 - "Students who study regularly perform better."
- A theoretical research approach asks:
 - What is "study regularly"?
 - How is performance measured?
 - What variables influence performance?
 - What mechanism explains the relationship?
 - Is the relationship causal?
 - Under what conditions does it occur?

12. Components of a Theory

- A theory commonly consists of:
- 12.1 Concepts
 - Abstract ideas representing phenomena.
 - Examples:
 - Intelligence
 - Motivation
 - Learning
 - Stress
 - Creativity
- 12.2 Constructs
 - A construct is a theoretically defined concept that may not be directly observable.
 - Examples:
 - Intelligence
 - Academic motivation
 - Creativity
 - Anxiety
 - Researchers operationalize constructs using measurable indicators.

- 12.3 Variables

- Variables are measurable characteristics that can take different values.
- Examples:
 - Age
 - Marks
 - Income
 - Study hours
 - Accuracy
 - Processing time

- 12.4 Propositions

- A proposition is a statement describing a relationship between concepts.
- Example:
 - Higher academic motivation is associated with greater learning engagement.

- 12.5 Hypotheses

- A hypothesis is a testable empirical statement derived from theory or prior evidence.
- Example:
 - Students using an adaptive learning system will achieve higher programming test scores than students receiving conventional instruction.

- 12.6 Relationships

- Theories specify how concepts or variables are related.
- For example:
 - Motivation → Study Engagement → Academic Performance

13. Functions of Theory

- Theory serves several important purposes.
- 13.1 Explanation
 - It explains why something happens.
- 13.2 Prediction
 - It helps predict what may happen under specified conditions.
- 13.3 Organization
 - It organizes existing knowledge into a coherent framework.
- 13.4 Guidance
 - It helps researchers:
 - Select variables
 - Formulate research questions
 - Develop hypotheses
 - Select methodologies
 - Interpret results
- 13.5 Identification of Research Gaps
 - Theory can reveal areas that have not yet been adequately investigated.
- 13.6 Generalization
 - A well-supported theory may help explain similar phenomena across different contexts.

14. Characteristics of a Good Theory

- A good scientific theory should generally be:
 1. Clear
 - Concepts should be understandable and appropriately defined.
 2. Logical
 - Relationships should be logically consistent.
 3. Testable
 - Its propositions should be capable of empirical examination where empirical testing is appropriate.
 4. Falsifiable
 - There should be possible evidence that could demonstrate that a proposition is incorrect.
 5. Consistent
 - It should not contain contradictory assumptions.
 6. Explanatory
 - It should provide meaningful explanations.
 7. Predictive
 - It should generate useful predictions where applicable.
 8. Parsimonious
 - It should explain phenomena without unnecessary complexity.
 9. Generalizable
 - Its explanatory principles should have applicability beyond a single observation when appropriate.
 10. Empirically supported
 - Scientific theories gain credibility through accumulated evidence.

15. Theory and Research: The Relationship

- Theory and research have a two-way relationship.
- Theory → Research
- Theory helps researchers:
 - Identify problems
 - Define concepts
 - Develop hypotheses
 - Select variables
 - Design studies
- Research → Theory
 - Research can:
 - Support theory
 - Refine theory
 - Extend theory
 - Challenge theory
 - Reject specific predictions
 - Generate new theory
- Thus:
- Theory → Hypothesis → Research → Evidence → Evaluation of Theory

16. Deductive Approach

- 16.1 Meaning
 - Deduction is reasoning that moves from general principles to specific conclusions.
 - Basic structure:
 - General $\rightarrow$ Specific
 - A researcher begins with:
 - Existing theory
 - General proposition
 - Hypothesis
 - Data collection
 - Testing
 - Conclusion

17. Example of Deductive Reasoning

- Suppose a theory states:
 - Increased practice improves skill performance.
- We develop the hypothesis:
 - Students who spend more time practicing programming will obtain higher programming scores.
- We then collect data.
- If the evidence supports the prediction, the hypothesis receives support.

- Structure

Theory



Proposition



Hypothesis



Data Collection



Analysis



Conclusion

18. Characteristics of Deductive Research

- Deductive research generally:
 - Begins with existing theory
 - Moves from general to specific
 - Uses hypotheses
 - Frequently uses quantitative methods
 - Tests relationships
 - Uses predefined variables
 - Often employs structured instruments
 - Uses statistical analysis
 - Seeks confirmation or disconfirmation of predictions

19. Example of Deductive Research

- Research question
 - Does the use of adaptive learning technology improve student performance?
- Theory
 - Adaptive learning theory suggests that personalized instruction can improve learning.
- Hypothesis
 - Students using an adaptive learning system will have significantly higher post-test scores than students using conventional learning materials.
- Method
 - Two groups are studied.
 - Experimental group → adaptive learning
 - Control group → conventional learning
- Analysis
 - Compare their post-test performance.
- Conclusion
 - Determine whether the observed evidence supports the hypothesis.

20. Inductive Approach

- 20.1 Meaning
 - Induction is reasoning that moves from specific observations to broader generalizations or theoretical explanations.
 - Basic structure:
 - Specific → General
 - The researcher begins with observations or data and identifies patterns.

21. Example of Inductive Reasoning

- Suppose a researcher interviews 50 teachers about generative AI.
- The researcher observes recurring themes:
 - AI saves preparation time.
 - AI helps generate examples.
 - AI can produce inaccurate information.
 - Teachers are concerned about academic integrity.
- The researcher develops broader propositions about teachers' perceptions of generative AI.

- Structure

Observation



Data



Patterns



Categories/Themes



Generalization



Theory or conceptual framework

22. Characteristics of Inductive Research

- Inductive research generally:
 - Begins with observations
 - Moves from specific to general
 - May not begin with a formal hypothesis
 - Identifies patterns
 - Generates concepts
 - Develops propositions
 - Can generate new theories
 - Frequently uses qualitative methods
 - Is often exploratory
- However, induction is not synonymous with qualitative research, and deduction is not synonymous with quantitative research. These associations are common but not absolute.

23. Example of Inductive Research

- Research problem
 - Little is known about how university teachers actually use generative AI in classroom preparation.
- Method
 - Conduct semi-structured interviews with teachers.
- Data
 - Interview transcripts.
- Analysis
 - Identify recurring themes.
 - Possible themes:
 - Lesson planning
 - Content generation
 - Assessment support
 - Time saving
 - Accuracy concerns
 - Academic integrity
- Outcome
 - Develop a conceptual framework explaining how teachers adopt and use generative AI.
- This is an inductive approach.

24. Deduction vs Induction

Feature	Deduction	Induction
Direction	General → Specific	Specific → General
Starting point	Theory	Observation/data
Main purpose	Test theory	Generate explanation/theory
Hypothesis	Usually explicit	May emerge during study
Nature	Confirmatory/testing	Exploratory/generative
Common methods	Quantitative	Often qualitative
Reasoning	Top-down	Bottom-up
Typical outcome	Support/refute prediction	Pattern/generalization/theory

25. Deductive and Inductive Reasoning: Simple Example

- Deductive

- All metals expand when heated.
- Iron is a metal.
- Therefore: Iron expands when heated.
- General principle → Specific conclusion

- Inductive

- Iron expands when heated.
- Copper expands when heated.
- Aluminium expands when heated.
- Gold expands when heated.
- Therefore: Metals generally expand when heated.
- Specific observations → General principle
- The inductive conclusion is probabilistic, not logically guaranteed.

26. Abduction

- Abduction is often described as:
 - Inference to the best explanation.
- It involves moving between observations, existing knowledge, hypotheses, and explanations.
- Example
- Observation:
 - Students who use a particular learning platform perform better.
- Possible explanations:
 - The platform improves learning.
 - More motivated students choose the platform.
 - Better students are more likely to use it.
 - Teachers using the platform may provide additional support.
- The researcher investigates which explanation best accounts for the evidence.

- Simple comparison
 - Deduction:
 - What should happen if the theory is correct?
 - Induction:
 - What general pattern emerges from observations?
 - Abduction:
 - What explanation best accounts for the observations?

27. Deductive–Inductive Research Cycle

In real research, deduction and induction often operate together.
A common research cycle is:

Existing Theory



Deductive Hypothesis



Empirical Investigation



Data



Unexpected Patterns



Inductive Interpretation



Theory Refinement



New Hypotheses



Further Testing

This makes research a **dynamic process rather than a one-directional activity.**

28. Scientific Method

- 28.1 Meaning
- The scientific method is a systematic approach for investigating phenomena, generating explanations, testing propositions, analyzing evidence, and developing reliable knowledge.
- In simplified form:
Observation → Problem → Question → Hypothesis → Investigation → Data → Analysis → Conclusion → Communication

The exact sequence can vary by discipline and research design.

29. Steps of the Scientific Method

- Step 1: Observation
 - The researcher notices a phenomenon.
 - Example: Students using an AI-based tutoring system appear to perform better.
- Step 2: Identification of Problem
 - The researcher identifies something that requires explanation.
 - Example:
 - It is unclear whether AI tutoring actually improves programming performance.
- Step 3: Research Question
 - The problem is converted into a focused question.
 - Does AI-based tutoring improve programming performance among postgraduate students?

- Step 4: Literature Review

- The researcher examines previous studies to determine:
 - What is already known?
 - What theories exist?
 - What methods have been used?
 - What findings have been reported?
 - What gaps remain?

- Step 5: Formulation of Hypothesis

- A testable prediction is developed.
- Example:
 - Students receiving AI-assisted tutoring will achieve higher programming test scores than students receiving conventional tutoring.

- Step 6: Research Design

- The researcher decides how to investigate the question.
- Possible designs:
 - Experimental
 - Quasi-experimental
 - Survey
 - Case study
 - Correlational
 - Longitudinal
 - Cross-sectional
 - Mixed methods

- Step 7: Data Collection

- Relevant evidence is collected.
- Examples:
 - Surveys
 - Interviews
 - Experiments
 - Observations
 - Databases
 - Sensor measurements
 - Documents

- Step 8: Data Analysis
 - Data are analyzed using appropriate techniques.
 - Examples:
 - Descriptive statistics
 - t-test
 - ANOVA
 - Regression
 - Correlation
 - Factor analysis
 - Thematic analysis
 - Content analysis
 - Machine learning methods
- Step 9: Interpretation
 - The researcher asks:
 - What do the findings mean?
 - Do they support the hypothesis?
 - Are alternative explanations possible?
 - Are the findings consistent with previous research?
- Step 10: Conclusion
 - The researcher provides an evidence-based answer to the research question.

- Step 11: Communication
 - Research findings are communicated through:
 - Research papers
 - Conferences
 - Theses
 - Reports
 - Datasets
 - Presentations
 - Technical documentation
- Step 12: Replication and Further Research
 - Other researchers may:
 - Replicate the study
 - Challenge the findings
 - Extend the research
 - Test it in another population
 - Use another methodology